A Cross-Species Lexical Repertoire Reveals a Biological Code Linking Acoustic Form to Communicative Meaning

Author One[1†⋆], Author Two[2†⋆], Author Three[1†], Author Four[2†],
Author Five[1], Author Six[1,2]

[1]Earth Species Project

[2]External Institution

Abstract

Understanding animal vocalizations means connecting acoustic form to communicative meaning. Years of field studies have produced extensive knowledge of such mappings, yet no unified framework exists for comparing them across species. We introduce AnimalLex, a curated cross-species repertoire database comprising 373 calls from 57 species spanning mammals, birds, and amphibians, annotated with standardized semantic and acoustic descriptions synthesized from the scientific literature and validated by domain experts. Using AnimalLex, we characterize what animals talk about and how. We find that acoustic and semantic distances are significantly correlated within species, across related species, and even across taxa—evidence for a conserved form-to-meaning structure that transcends phylogenetic boundaries. We leverage this structure to perform cross-species lexical translation: given any call, we can identify its closest analogue in any other species. Projecting these into human language, we can also identify the closest semantic analogue in our own vocabulary for any documented call. These results suggest that a biological code linking acoustic form to communicative function is a deep organizational principle of animal communication, and that large-scale repertoire curation—now made practical by language models—unlocks new possibilities for comparative and translational bioacoustics. An interactive tool is provided to explore the database and perform translations at scale.

1 Introduction

Decades of field research have established that animal vocalizations are not simple reflexes but structured communicative signals whose forms are shaped by function Marler, 1955. Alarm calls are short, broadband, and easily localized—suited for rapid threat broadcast. Contact calls are tonal and repetitive—suited for long-distance propagation and individual recognition. Infant-directed calls are high-pitched across taxa—plausibly linked to shared attentional mechanisms. Yet these patterns have been documented one species at a time, and no synthesis has asked whether they constitute a universal code: a systematic, learnable relationship between acoustic form and communicative meaning that holds across the tree of life.

The question is not merely empirical but conceptual. In linguistics, the arbitrariness of the sign—the principle that a word's sound bears no necessary relation to its meaning—has been a foundational tenet since Saussure Saussure, 1916. Counter-evidence is accumulating: sound symbolism and crosslinguistic iconicity demonstrate that form-to-meaning mappings

⋆Corresponding author: [PLACEHOLDER]@earthspecies.org
†These authors contributed equally to this work

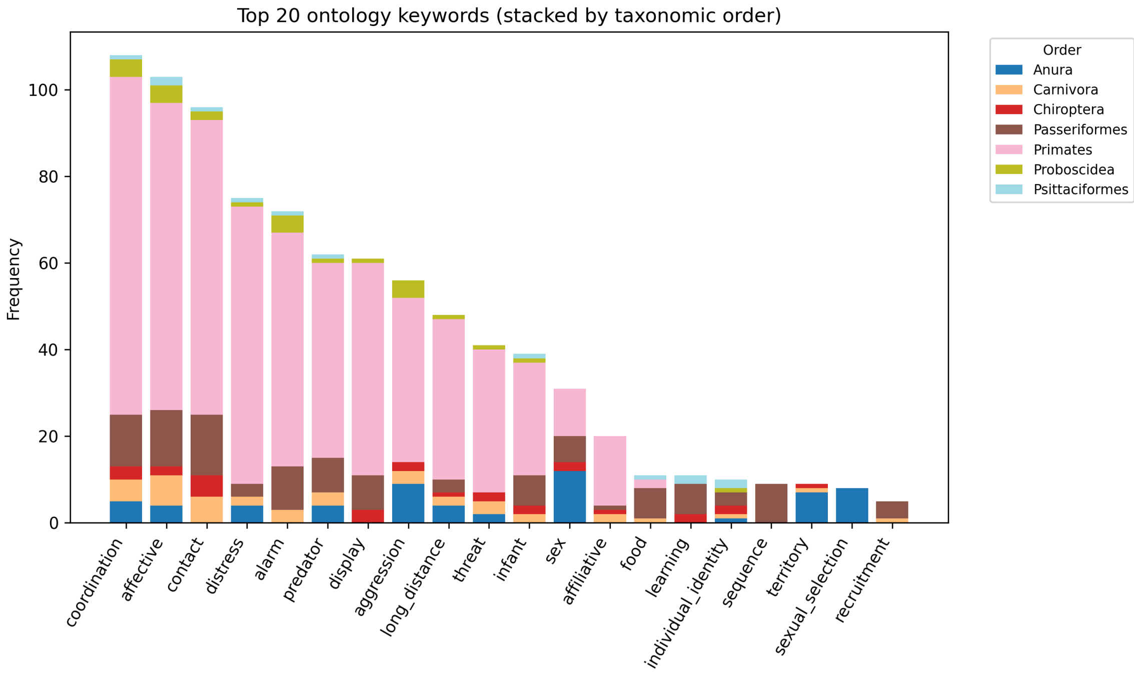

Figure 1. **AnimalLex: a curated cross-species lexical repertoire.** **(A)** Frequency of semantic ontology keywords across all 373 calls, colored by broad communicative category. Coordination, contact, alarm, and affective signaling account for more than half of all calls; rarer categories include referential signaling, individual identity coding, and learning-related vocalizations. **(B)** [Placeholder: distribution of acoustic keywords extracted from free-text descriptions, showing the prevalence of spectro-temporal features such as tonal, broadband, high-frequency, pulsed, and frequency-modulated.] **(C)** Taxonomic breakdown of the 57 species in AnimalLex across three vertebrate classes (Mammalia, Aves, Amphibia). **(D)** Calls per species, ranging from 3 to 12+. Repertoire size reflects both biological diversity and the depth of available primary literature.

exist and are learned in human language Dingemanse et al., 2015. In non-human animals the question is more open still. If such a biological code exists, it could arise through two mechanisms: convergent evolution driven by shared physical and ecological constraints (e.g., the localizability demands of alarm signaling), or phylogenetic conservation of ancestral form-to-meaning associations within lineages. These mechanisms make different predictions—the former produces cross-taxon universals, the latter family-level regularities—and can in principle be distinguished with sufficient data.

Here we test for a biological code at scale by assembling AnimalLex, a curated database of 373 calls from 57 species (Fig. 1). Each entry pairs a free-text acoustic description with a free-text semantic description and standardized ontology keywords, drawn from peer-reviewed literature and validated by domain experts. We embed both descriptions independently into continuous vector spaces and ask whether their pairwise distance structures are aligned—within species, across species, and across higher taxa. We find that they are: acoustic distance is a significant predictor of semantic distance at every taxonomic level tested. A simple linear model trained on this alignment generalizes across species and taxa, demonstrating that the code is learnable. We apply it directly to perform cross-species lexical translation—mapping any documented call to its closest human-language analogue, and any human concept to its closest animal vocalization—and provide an interactive web application for broad exploration.

2 A cross-species lexical repertoire

AnimalLex was assembled by prompting a large language model (GPT-4o) to structure knowledge from peer-reviewed literature for each target species, producing entries that record call name, free-text acoustic description, free-text semantic description, standardized ontology

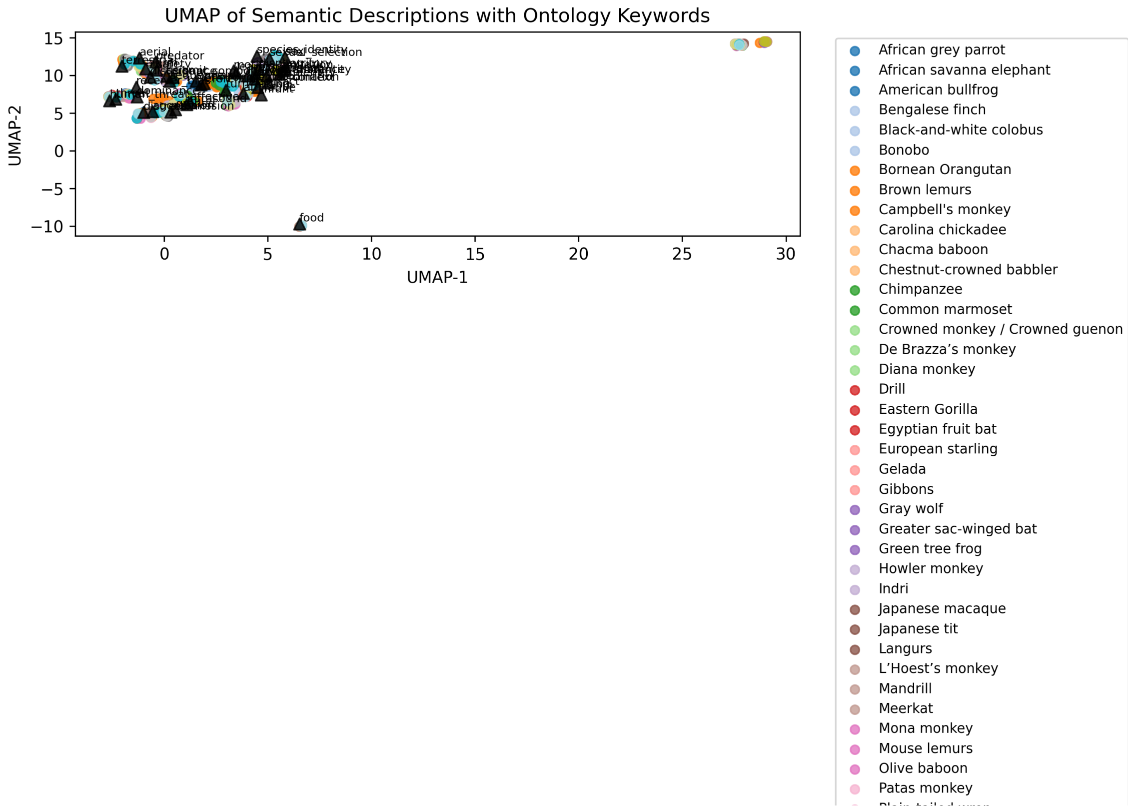

Figure 2. **Alignment of acoustic and semantic embedding spaces. (A, B)** UMAP projections of semantic (A) and acoustic (B) embeddings of all 373 calls in AnimalLex, colored by ontology keyword. Calls with related communicative functions cluster in semantic space; calls described in similar acoustic terms cluster in acoustic space. **(C)** [Placeholder: Mantel scatter plot—pairwise acoustic distance vs. pairwise semantic distance for all call pairs, colored by taxonomic relationship (within-species, same family, across families). Points above the diagonal are alarm/distress calls with matched acoustic profiles; points to the right are acoustically similar contact calls with shared semantic labels. Mantel r and p values annotated for each group.] The correspondence between panels (A) and (B)—similar topologies, similar cluster identities—is the visual signature of the biological code quantified in the main text.

keywords, subjective reliability, and primary references. Entries were then reviewed and corrected by domain experts (see Methods). The 57 species span three vertebrate classes: 43 mammals, 9 birds, and 5 amphibians, covering taxa from African savanna elephants (*Loxodonta africana*) and gray wolves (*Canis lupus*) to Carolina chickadees (*Poecile carolinensis*) and túngara frogs (*Engystomops pustulosus*).

The most common semantic categories are coordination ($n = 106$), affective signaling ($n = 98$), contact and cohesion ($n = 94$), distress ($n = 72$), and alarm ($n = 71$; Fig. 1A). Less frequent but theoretically important categories include referential signals ($n = 5$), individual-identity calls ($n = 8$), and learning-associated vocalizations ($n = 10$). On the acoustic side, descriptions span the full range of spectro-temporal parameters: tonal vs. broadband energy, frequency range and modulation, temporal patterning (pulsed, repetitive, graded), and call-internal structure (notes, syllables, phrases; Fig. 1B). Together these two annotation axes provide the substrate for testing form-to-meaning alignment.

3 Acoustic and semantic spaces are aligned across the tree of life

We embedded each call’s acoustic and semantic descriptions separately into a shared sentence-transformer space (see Methods) and computed pairwise cosine similarities across all $373(373-1)/2 = 69{,}378$ call pairs. A Mantel test Mantel, 1967 of the correlation between acoustic and semantic similarity matrices yields $r = 0.31$, $p < 0.001$ (9,999 permutations; Fig. 2C). The effect is

Table 1. **Cross-generalization of acoustic-to-semantic prediction.** Four models are evaluated under three hold-out conditions of increasing difficulty. *Cos.* = mean cosine similarity between predicted and true semantic embedding (higher is better, max 1). *MRR* = mean reciprocal rank of the true call among all test-set calls ranked by similarity to the prediction (higher is better, max 1). The random baseline shuffles acoustic–semantic pairings within each test set. Retrieval ($k = 1$) returns the nearest neighbour in acoustic space among training calls. Within-repertoire values are means over 10 random folds; held-out species and family values are means over leave-one-out folds (55 and 26 folds respectively).

	Within repertoire		**Held-out species**		**Held-out family**	
Model	Cos.	MRR	Cos.	MRR	Cos.	MRR
Random	0.36	0.11	0.46	0.41	0.47	0.42
Retrieval ($k = 1$)	0.67	0.51	0.68	0.78	0.60	0.67
Linear (ridge)	**0.74**	0.46	**0.74**	**0.80**	**0.68**	**0.70**
MLP	0.71	0.45	0.73	0.80	0.66	0.68

present at every taxonomic level: within individual species ($r = 0.23$, $p < 0.001$), across species within the same family ($r = 0.49$, $p < 0.001$), and across species from different families ($r = 0.23$, $p < 0.001$). Strikingly, the correlation is strongest for same-family cross-species pairs, suggesting that acoustic-semantic alignment is most visible when species share evolutionary history but have begun to acoustically diverge—a pattern consistent with the biological code being phylogenetically conserved while acoustic form drifts.

To decompose the signal into interpretable units, we computed the pointwise mutual information (PMI) between acoustic and semantic ontology keywords across all calls (Fig. 3). High-frequency and broadband acoustic features associate specifically with alarm and threat semantics—consistent with the localizability demands of anti-predator signaling Marler, 1955. Tonal and repetitive features associate with contact and coordination semantics—consistent with the propagation demands of long-range cohesion calls. High-pitched features associate with infant-directed and distress calls—consistent with cross-species patterns in arousal-linked vocalizations Ohala, 1994. These keyword-level associations provide a transparent, mechanistic decomposition of the form-to-meaning code.

4 Form-to-meaning mapping is learnable and generalizes across taxa

The previous section demonstrates that there is stability in the form-to-meaning mapping. We trained models to exploit this regularity: predicting semantic embeddings from acoustic embeddings under three hold-out conditions of increasing ambition (Table 1). Within-repertoire, ridge regression achieves a mean cosine similarity of 0.74 and MRR of 0.46—substantially above the random baseline (cos. 0.36, MRR 0.11). Generalization to held-out species is equally strong (cos. 0.74, MRR 0.80), and remains well above chance even when entire families are held out (cos. 0.68, MRR 0.70). The retrieval baseline approaches ridge regression in MRR for held-out conditions, indicating that the acoustic-to-semantic mapping is largely a global alignment of the two spaces rather than a complex nonlinear transformation. The MLP offers no consistent advantage over ridge regression, consistent with this interpretation. Together these results show that the biological code identified by the Mantel analysis is not an artefact of within-species structure: it generalises to species and entire families unseen at training time. In other words, our existing knowledge of species transfers to new species.

5 Lexical translation across species and into human language

The alignment between acoustic and semantic spaces supports a direct translational application. Given any human concept, we embed it in the same semantic space and retrieve the call in AnimalLex with the nearest semantic embedding. Conversely, given any documented call, we retrieve the closest human concept. Table 2 and Fig. 4 show ten examples: the translations

Table 2. **Lexical translation: selected examples.** For each human concept (left), the best-matching call in AnimalLex by semantic embedding similarity is listed with its cosine similarity score, species, and a condensed acoustic description. [Placeholder: to be filled once translation pipeline is run.]

Human concept	**Sim.**	**Best-match call**	**Acoustic description (condensed)**
danger	—	[species] / [call]	[acoustic]
I am hungry	—	[species] / [call]	[acoustic]
come here	—	[species] / [call]	[acoustic]
I am here	—	[species] / [call]	[acoustic]
stay away	—	[species] / [call]	[acoustic]
greeting	—	[species] / [call]	[acoustic]
let's move together	—	[species] / [call]	[acoustic]
I am afraid	—	[species] / [call]	[acoustic]
this is mine	—	[species] / [call]	[acoustic]
help me	—	[species] / [call]	[acoustic]

are intuitive—"danger" maps to anti-predator alarm calls; "come here" maps to contact recruitment calls; "I am hungry" maps to begging vocalizations—and are grounded in the entire corpus of documented animal vocalizations as synthesized by AnimalLex.

This retrieval pipeline is necessarily limited to calls that have been documented and annotated. We extend it using the acoustic-to-semantic predictor from the previous section: for any acoustic recording from any species—including taxa not represented in AnimalLex—the model provides a best-guess semantic embedding from acoustic features alone, enabling translation beyond the boundaries of existing knowledge. We provide an interactive web application (see Supplementary) through which both modes of translation can be explored at scale.

6 Discussion

Our results establish three things. First, animal vocalizations carry systematic form-to-meaning relationships that are detectable across the tree of life—not just within well-studied model systems. Second, these relationships are learnable and generalizable: a model trained on some species predicts meaning from acoustics for held-out species and families. Third, this learnability supports practical translation between acoustic and semantic spaces, including across the human-animal boundary.

Two non-exclusive mechanisms can explain the cross-taxon alignment. Convergent functional pressures may independently channel evolution toward similar acoustic solutions for similar communicative problems: the physics of sound localization favors broadband calls for alarm regardless of ancestry Marler, 1955; the acoustics of long-range propagation favor tonality for cohesion calls Morton, 1977. Phylogenetic inertia may additionally preserve ancestral form-to-meaning associations within lineages, explaining why correlations are strongest within species and weaken—without vanishing—across families. These mechanisms predict different patterns in phylogenetically controlled analyses, which future work with broader taxonomic sampling can test.

Several caveats apply. AnimalLex represents the current state of published knowledge, which skews toward well-studied mammals; birds and amphibians—acoustically diverse and experimentally tractable—are underrepresented. Our acoustic embeddings are derived from textual descriptions, not from audio directly; future work integrating audio representations Hagiwara, 2022; Robinson et al., 2025 will test whether the form-to-meaning alignment is even stronger in the raw signal domain. Finally, lexical translation should be understood as semantic proximity, not cognitive equivalence: when we say that "danger" maps to an alarm call, we make no claim about the mental states of the signaling animal.

Despite these limitations, the demonstration that form-to-meaning structure is learnable and cross-taxon generalizable opens a practical path forward: with each new species added to AnimalLex, the translation model improves, and with each improvement the model can bootstrap annotation of the next species. This virtuous cycle—human knowledge, compressed by language models, refined by acoustic-semantic prediction, extended to undocumented taxa—is the long-term promise of the approach introduced here.

7 Methods

7.1 Database construction

AnimalLex was assembled in two stages. We first prompted GPT-4o OpenAI, 2024 with a structured schema specifying six required fields per call: (1) call name, (2) free-text acoustic description (spectro-temporal features, amplitude envelope, call-internal structure), (3) free-text semantic description (communicative function, behavioral context, receiver responses), (4) ontology keywords drawn from a fixed controlled vocabulary, (5) subjective reliability rating (high / medium / low / contested), and (6) URLs to primary-literature references. The model was instructed to restrict outputs to findings supported by peer-reviewed sources and to flag contested or uncertain claims. In the second stage, domain experts in avian, mammalian, and amphibian bioacoustics reviewed all entries, corrected factual errors, added missing calls, and updated reliability ratings. Calls rated "low" or "contested" are retained in AnimalLex but excluded from quantitative analyses reported here.

7.2 Embeddings

Acoustic and semantic descriptions were embedded independently using the all-MiniLM-L6-v2 sentence transformer Reimers and Gurevych, 2019, which maps variable-length text to a 384-dimensional vector space trained on semantic similarity tasks. Embeddings were ℓ_2-normalized before computing cosine distances. We deliberately used textual embeddings rather than audio-derived representations to ensure that acoustic and semantic descriptions were embedded in a comparable space; the validity of this choice is discussed above and tested in Supplementary B.

7.3 Mantel test

For each subset of calls (all pairs; within-species pairs; within-family pairs; cross-family pairs), we computed the Pearson correlation between the upper triangle of the pairwise acoustic distance matrix and the upper triangle of the pairwise semantic distance matrix. Statistical significance was assessed by permuting the rows and columns of the acoustic distance matrix 10,000 times and computing the fraction of permuted correlations exceeding the observed value.

7.4 PMI analysis

Acoustic keywords were extracted from free-text acoustic descriptions by matching a fixed lexicon of 20 spectro-temporal terms (e.g., *tonal*, *broadband*, *high-frequency*). Semantic keywords were taken directly from the ontology keyword field. For each acoustic–semantic keyword pair (a, s), PMI was estimated as $\log_2 \hat{p}(a, s) - \log_2 \hat{p}(a) - \log_2 \hat{p}(s)$, where probabilities were estimated from call frequencies. Pairs with fewer than three co-occurrences were excluded.

7.5 Acoustic-to-semantic prediction

We trained a ridge regression mapping acoustic embeddings $\mathbf{x} \in \mathbb{R}^{384}$ to semantic embeddings $\mathbf{y} \in \mathbb{R}^{384}$, regularized with ℓ_2 penalty λ selected by leave-one-species-out cross-validation. Predicted embeddings were ℓ_2-normalized before evaluation. Performance was measured by (i) mean cosine similarity between predicted and true semantic embeddings and (ii) mean reciprocal rank (MRR) of the true call's semantic embedding among all held-out calls, ranked by cosine similarity to the prediction. We additionally evaluated a shallow MLP (two hidden layers of 512 units, ReLU activations, dropout 0.2) and a retrieval baseline (k-nearest neighbors in acoustic space, semantic embedding averaged over the k neighbors). Cross-validation used 10 folds stratified at the species level.

8 Contributions

Conceptualization: A.B., C.D. **Methodology:** A.B., C.D., E.F. **Software:** E.F., G.H. **Validation:** C.D., E.F. **Formal analysis:** A.B., E.F. **Investigation:** A.B., C.D. **Resources:** I.J. **Data curation:** A.B., C.D., E.F. **Writing – original draft:** A.B. **Writing – review & editing:** A.B., C.D., E.F., I.J. **Visualization:** E.F., G.H. **Supervision:** I.J. **Project administration:** I.J. **Funding acquisition:** I.J.

9 Acknowledgements

We thank the researchers whose detailed ethological and bioacoustic studies form the foundation of AnimalLex.

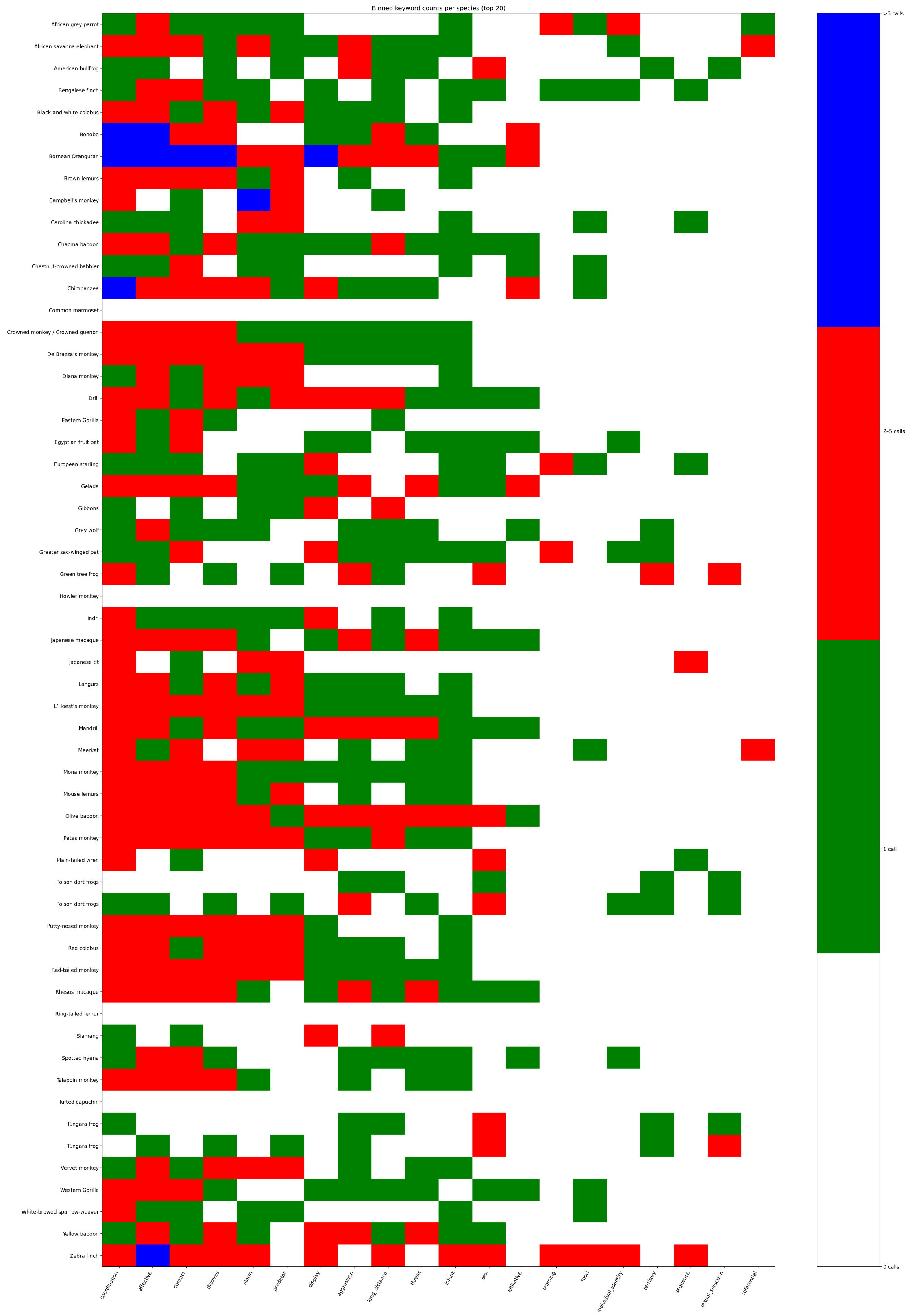

Figure 3. **Keyword-level structure of the biological code.** PMI between acoustic keywords (rows) and semantic keywords (columns), estimated across all 373 calls in AnimalLex. Warm colors indicate co-occurrence above chance; cool colors indicate anti-association. The strongest positive associations—high-frequency/broadband × alarm/predator; tonal/repetitive × contact/coordination; high-pitched × infant/distress—are consistent with functional acoustic ecology hypotheses and extend them across taxa.

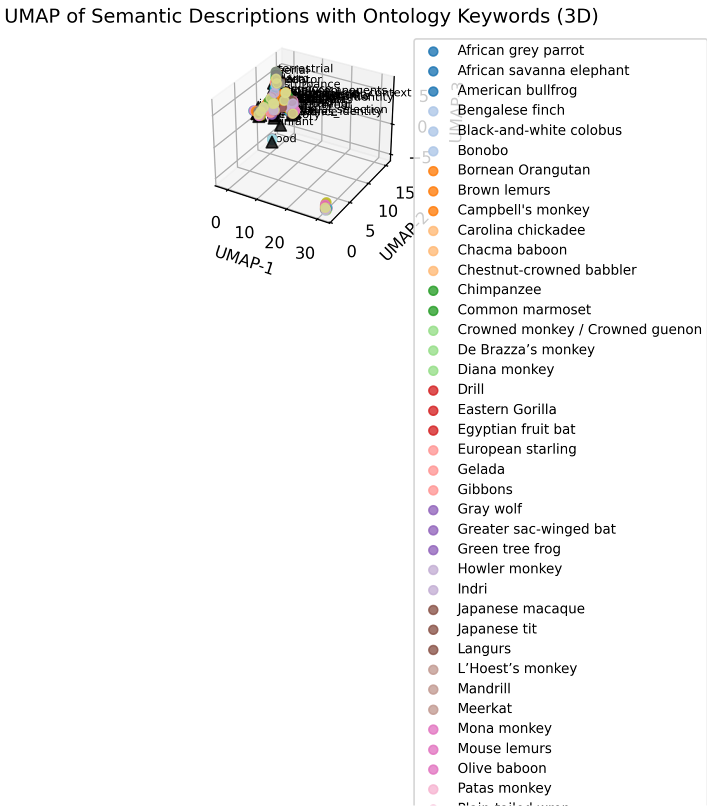

Figure 4. **Cross-species lexical translation.** [Placeholder: For each of ten human concepts (left column), the best-matching animal call in AnimalLex is identified by semantic embedding proximity. Lines connect each concept to its translation; line thickness encodes cosine similarity. Inset spectrograms (or schematic waveform icons) illustrate the acoustic character of each best-matching call. Color encodes taxonomic class of the matched species. The diversity of matched species and the intuitiveness of many translations—"danger" → meerkat alarm call; "come here" → elephant contact rumble—illustrate the reach and coherence of the form-to-meaning code.]

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A Web application

We release an interactive web application at [placeholderURL] that allows users to: (i) browse AnimalLex filtered by species, taxon, semantic category, or acoustic keyword; (ii) perform lexical translation—enter any text and retrieve the closest animal calls by semantic similarity; (iii) perform reverse translation—select any call and retrieve the closest human-language descriptions; and (iv) explore semantic and acoustic embedding spaces as interactive 3-D UMAP projections. The application is built on [placeholder: framework] and is freely accessible without registration.

B Validation against audio representations

[Placeholder: comparison of Mantel test results when acoustic embeddings are derived from (a) text descriptions, as in the main analysis, vs. (b) audio features extracted by AVES Hagiwara, 2022 or NatureLM-audio Robinson et al., 2025 for the subset of calls for which verified audio samples are available. This analysis tests whether the form-to-meaning alignment detected through textual descriptions is an artifact of description style or reflects genuine acoustic structure.]

C Full database statistics

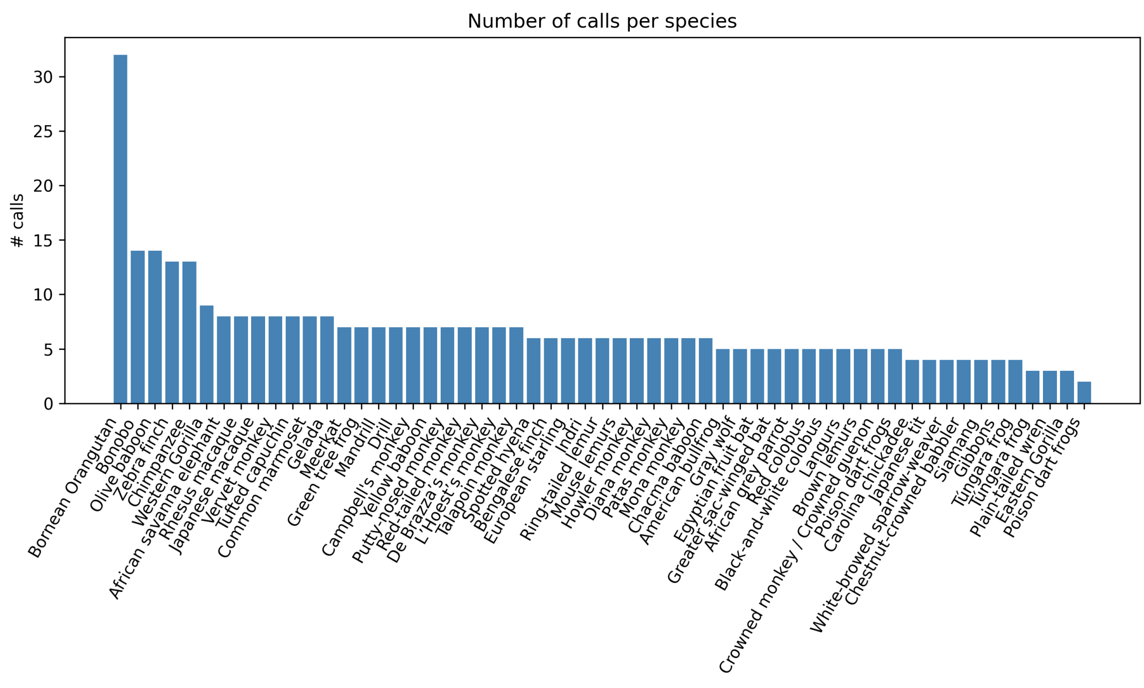

Figure 5. **Calls per species in AnimalLex**, sorted by count. Color encodes taxonomic class. The median repertoire size is [X] calls; the largest repertoire (*Loxodonta africana*) contains [X] calls.

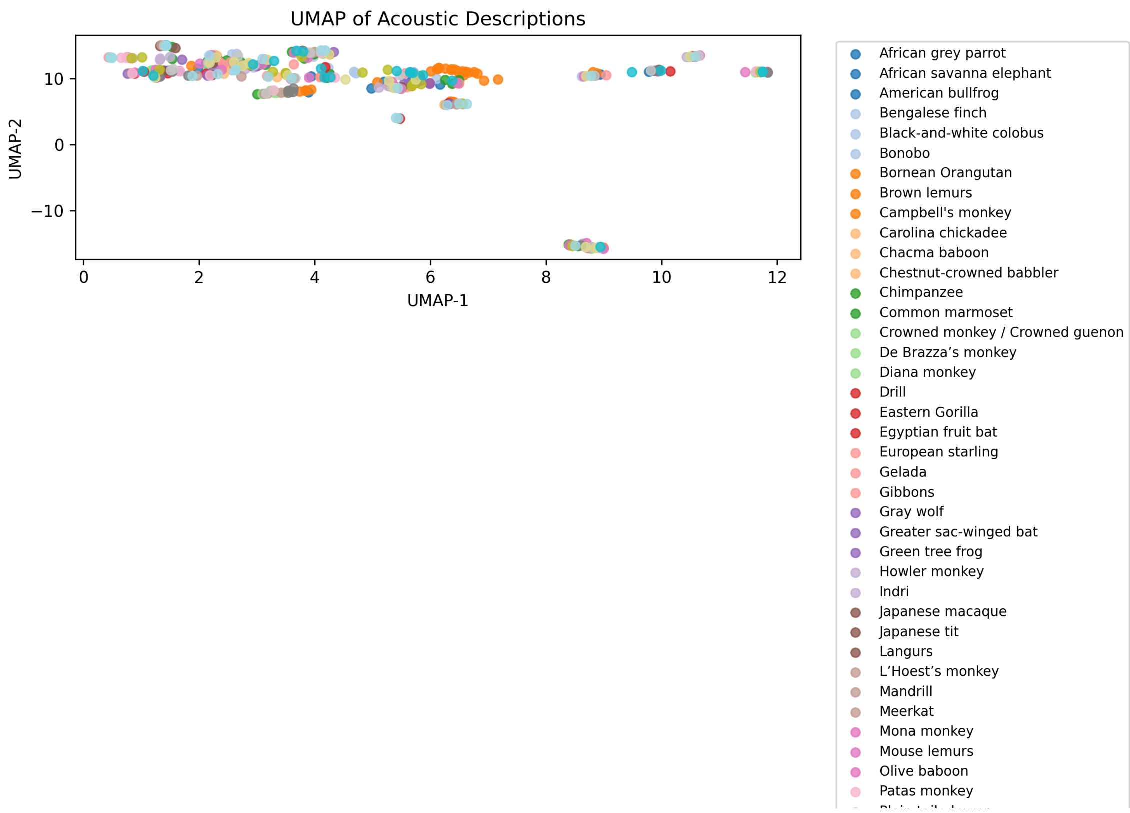

Figure 6. **Acoustic embedding space (UMAP).** Each point is a call in AnimalLex, positioned by acoustic description embedding and colored by taxonomic class. Clusters correspond to spectro-temporal call types (tonal long calls, broadband alarm bursts, etc.) that recur across classes, illustrating the convergent acoustic solutions captured by the database.